they were physically present. But things are much trickier when physical space is augmented or replaced by cyberspace and when more and more transactions involve only the transfer of information rather than of physical goods or traditional currencies. For example, a citizen of Uruguay may daily interact online with numerous companies that might have no physical presence in Uruguay but that provide her with various services. Google provides her with free search, and ByteDance – the parent company of the TikTok application – provides her with free social media. Other foreign companies routinely target her with advertisements: Nike wants to sell her shoes, Peugeot wants to sell her a car, and Coca-Cola wants to sell her soft drinks. In order to target her, these companies buy both personal information and ad space from Google and ByteDance. In addition, Google and ByteDance use the information they harvest from her and from millions of other users to develop powerful new AI systems that they can then sell to various governments and corporations throughout the world. Thanks to such transactions, Google and ByteDance are among the richest corporations in the world. So, should her transactions with them be taxed in Uruguay?

Some think they should. Not just because information from Uruguay helped make these corporations rich, but also because their activities undermine taxpaying Uruguayan businesses. Local newspapers, TV stations and cinemas lose customers and ad revenue to the tech giants. Prospective Uruguayan AI companies also suffer, because they cannot compete with Google' and ByteDance's massive data troves. But the tech giants reply that none of the relevant transactions involved any physical presence in Uruguay or any monetary payments. Google and ByteDance provided Uruguayan citizens with free online services, and in return the citizens freely handed over their purchase histories, vacation photos, funny cat videos and other information.

If they nevertheless want to tax these transactions, the tax authorities need to reconsider some of their most fundamental concepts, such as 'nexus'. In tax literature, 'nexus' means an entity's connection to a given jurisdiction. Traditionally, whether a corporation had nexus in a specific country depended on whether it had physical presence there, in the form of offices, research centres, shops and so forth. One proposal for addressing the tax dilemmas created by the computer network is to redefine nexus. In the words of the economist Marko Köthenbürger, 'The definition of nexus based on a physical presence should be adjusted to include the notion of a